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BSCS 3-A
CS PROFESSIONAL ELECTIVE

Model Context Protocol - client-server protocol, basically it standardizes instructions for LLMs/agents on how to communicate with external tools such as Github or Gmail through structured JSON contracts.

RESOURCES USED:

GENERATIVE AI TOOLS:

- OPENAI(CHATGPT & CODEX) - PLANNING AND IMPLEMENTATION
- ANTROPHIC(CLAUDE) - DEMO AND TESTING

OFFICIAL DOCUMENTATIONS USED:

- MCP TUTORIAL
- Claude connector documentation

Technology	Use
Node.js + TypeScript	Build and run the server
MCP TypeScript SDK	Register tools and handle MCP requests
Zod	Input schema, Validate tool inputs
Express + Streamable HTTP	Serve the /mcp endpoint
ngrok	Expose the server through HTTPS
PM2	Manages the server
Git	Version control and change inspection
oidc-provider	Issues token, OAuth authentication for Claude

OAuth with CIMD to connect MCP server to Claude Web via connectors

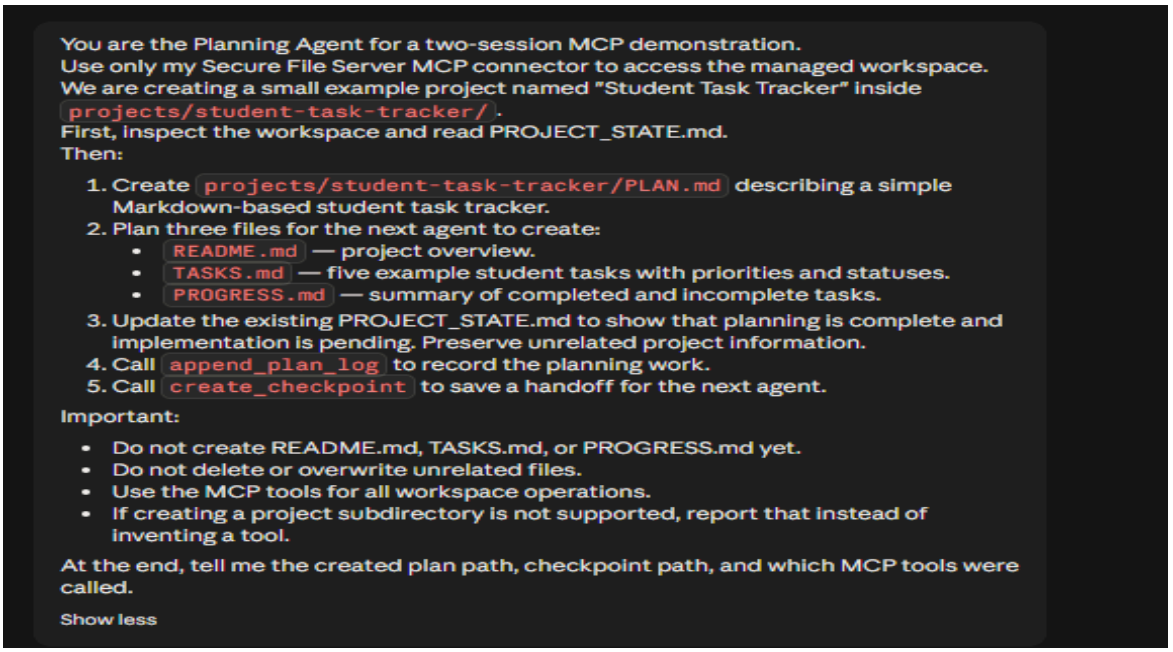
Tool	Function
list_files	List permitted files and folders in a directory
read_file	Read a text file
write_file	Create a text file without overwriting an existing file
str_replace	Replace one matching text occurrence using the current file hash
delete_file	Move a file to recovery storage with confirmation
search_files	Search permitted filenames or text contents
git_status	Show workspace Git changes
git_diff	Show changes to tracked files
append_plan_log	Append a structured progress entry
create_checkpoint	Save a handoff summary linked to the current project state

Claude sends requests through ngrok to the local MCP server. The server authenticates each request, validates its inputs, and performs permitted operations inside the configured workspace. Handoff files preserve context between sessions.

- **M1 — Skeleton:** Set up the TypeScript MCP server and verified it could expose a basic tool.

- **M2 — Sandbox:** Added the file tools and workspace path protections.
- **M3 — Deployment:** Hosted the server locally with PM2 and exposed it over HTTPS through ngrok, with bearer-token authentication.
- **M4 — Handoff and Git:** Added Git inspection tools and workspace files for plans, decisions, and checkpoints.
- **M5 — Demonstration:** Connected Claude Web and used two separate chats to plan a Student Task Tracker in one session, then continue it in another using the saved handoff files.

AGENT 1:



You are the Planning Agent for a two-session MCP demonstration.
Use only my Secure File Server MCP connector to access the managed workspace.
We are creating a small example project named "Student Task Tracker" inside `projects/student-task-tracker/`.
First, inspect the workspace and read PROJECT_STATE.md.
Then:

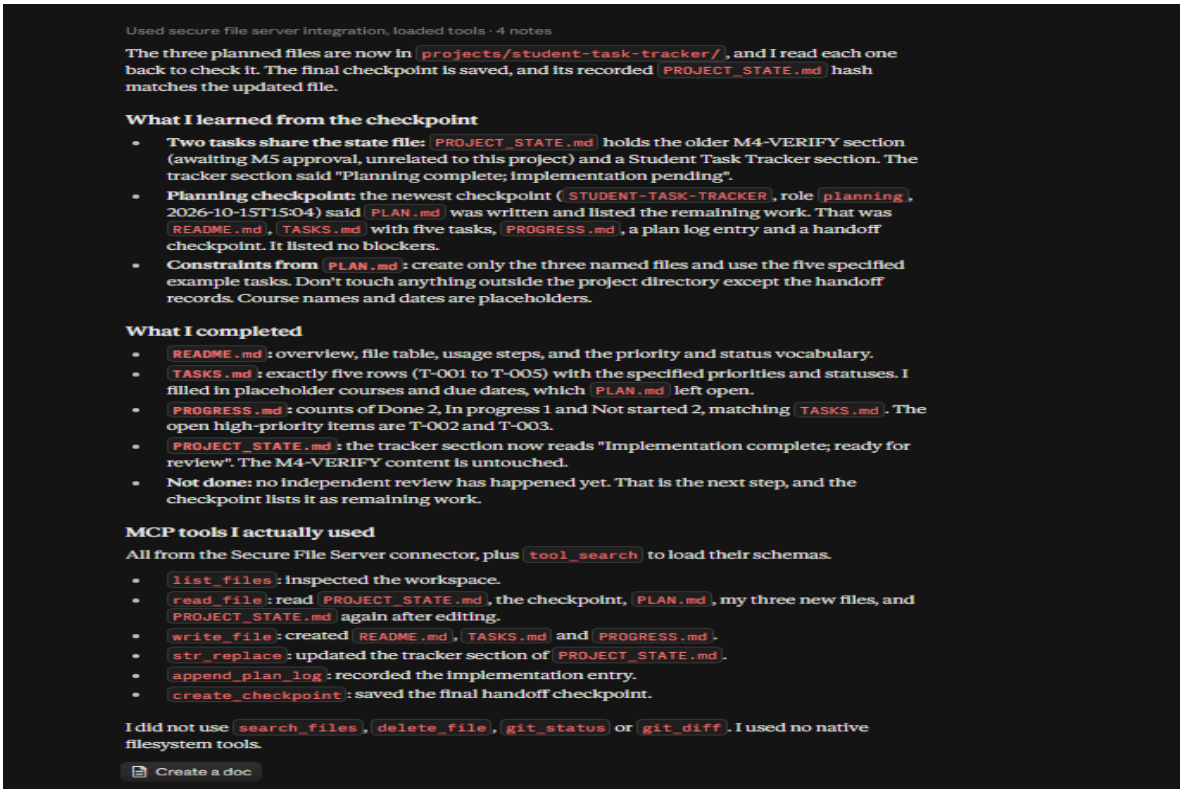
1. Create `projects/student-task-tracker/PLAN.md` describing a simple Markdown-based student task tracker.
2. Plan three files for the next agent to create:
   - `README.md` — project overview.
   - `TASKS.md` — five example student tasks with priorities and statuses.
   - `PROGRESS.md` — summary of completed and incomplete tasks.
3. Update the existing PROJECT_STATE.md to show that planning is complete and implementation is pending. Preserve unrelated project information.
4. Call `append_plan_log` to record the planning work.
5. Call `create_checkpoint` to save a handoff for the next agent.

Important:

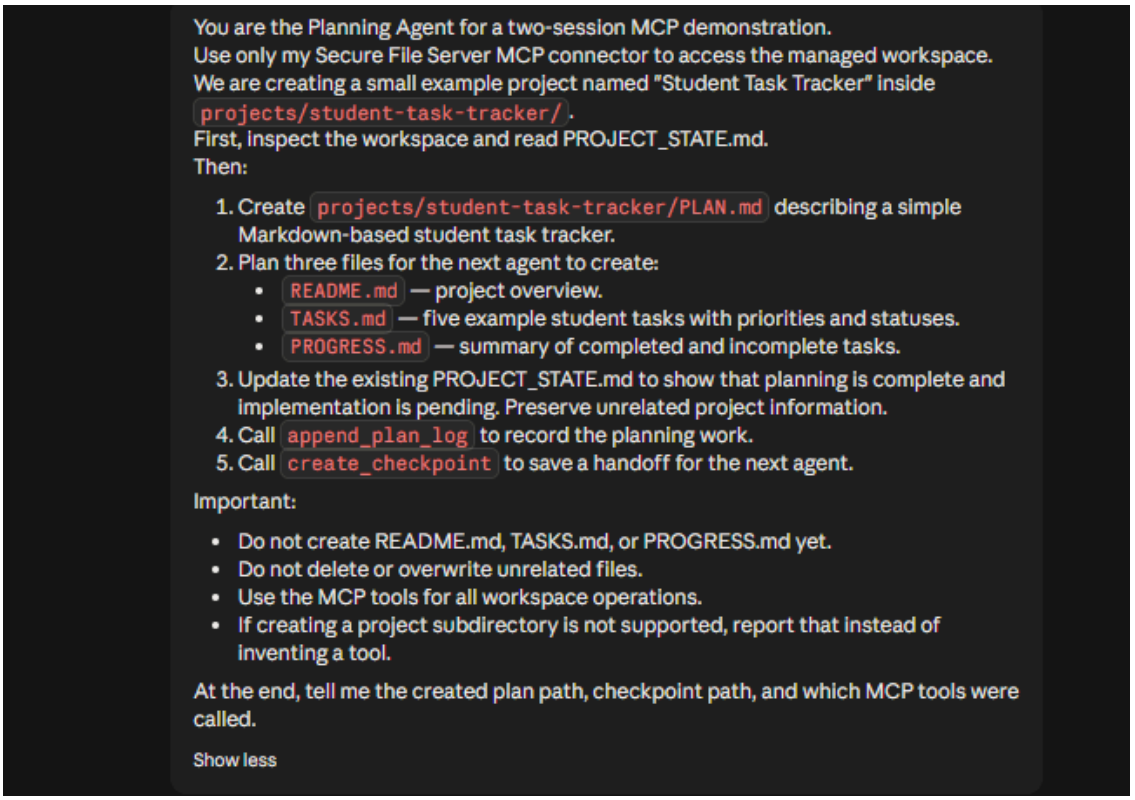
- Do not create README.md, TASKS.md, or PROGRESS.md yet.
- Do not delete or overwrite unrelated files.
- Use the MCP tools for all workspace operations.
- If creating a project subdirectory is not supported, report that instead of inventing a tool.

At the end, tell me the created plan path, checkpoint path, and which MCP tools were called.

Show less


Used secure file server integration, loaded tools · 4 notes
The three planned files are now in projects/student-task-tracker/, and I read each one back to check it. The final checkpoint is saved, and its recorded PROJECT_STATE.md hash matches the updated file.
What I learned from the checkpoint
Two tasks share the state file: PROJECT_STATE.md holds the older M4-VERIFY section (awaiting M5 approval, unrelated to this project) and a Student Task Tracker section. The tracker section said "Planning complete; implementation pending".
Planning checkpoint: the newest checkpoint (STUDENT-TASK-TRACKER, role planning, 2026-10-15T15:04) said PLAN.md was written and listed the remaining work. That was README.md, TASKS.md with five tasks, PROGRESS.md, a plan log entry and a handoff checkpoint. It listed no blockers.
Constraints from PLAN.md: create only the three named files and use the five specified example tasks. Don't touch anything outside the project directory except the handoff records. Course names and dates are placeholders.
What I completed
README.md: overview, file table, usage steps, and the priority and status vocabulary.
TASKS.md: exactly five rows (T-001 to T-005) with the specified priorities and statuses. I filled in placeholder courses and due dates, which PLAN.md left open.
PROGRESS.md: counts of Done 2, In progress 1 and Not started 2, matching TASKS.md. The open high-priority items are T-002 and T-003.
PROJECT_STATE.md: the tracker section now reads "Implementation complete; ready for review". The M4-VERIFY content is untouched.
Not done: no independent review has happened yet. That is the next step, and the checkpoint lists it as remaining work.
MCP tools I actually used
All from the Secure File Server connector, plus tool_search to load their schemas.
list_files: inspected the workspace.
read_file: read PROJECT_STATE.md, the checkpoint, PLAN.md, my three new files, and PROJECT_STATE.md again after editing.
write_file: created README.md, TASKS.md and PROGRESS.md.
str_replace: updated the tracker section of PROJECT_STATE.md.
append_plan_log: recorded the implementation entry.
create_checkpoint: saved the final handoff checkpoint.
I did not use search_files, delete_file, git_status or git_diff. I used no native filesystem tools.
Create a doc

AGENT 2:


You are the Planning Agent for a two-session MCP demonstration.
Use only my Secure File Server MCP connector to access the managed workspace.
We are creating a small example project named "Student Task Tracker" inside projects/student-task-tracker/.
First, inspect the workspace and read PROJECT_STATE.md.
Then:
1. Create projects/student-task-tracker/PLAN.md describing a simple Markdown-based student task tracker.
2. Plan three files for the next agent to create:
README.md — project overview.
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3. Update the existing PROJECT_STATE.md to show that planning is complete and implementation is pending. Preserve unrelated project information.
4. Call append_plan_log to record the planning work.
5. Call create_checkpoint to save a handoff for the next agent.
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Do not delete or overwrite unrelated files.
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